

Arduino Smart Elderly Alert System - Case Study

Overview

This project is a smart alert system prototype for elderly safety. It uses an ultrasonic sensor to monitor movement and triggers staged alerts using an LED, servo motor, and buzzer when extended inactivity is detected.

Problem

Elderly individuals may struggle to call for help during emergencies, especially if they fall unconscious or are unable to move. The system explores how a simple embedded prototype can provide both automatic and manual emergency alerts.

Hardware

- Arduino-compatible microcontroller
- Ultrasonic distance sensor
- Push button
- Buzzer
- LED
- Servo motor

Alert Logic

The system escalates through three demo stages: LED warning after 5 seconds, servo nudge after 10 seconds, and full emergency mode after 15 seconds without movement. These short thresholds were used for demonstration and should be extended in real deployment.

Key Learning

The project strengthened practical skills in sensor integration, Arduino C++ programming, actuator control, timing logic, and designing assistive technology with false alarm reduction in mind.